Limiting & Excess Reagents

In the synthesis of water, we require *exactly* 2 mol of H_2 and *exactly* 1 mol O_2 to create *exactly* 2 mol H_2O

$$2\ H_{2(g)} + O_{2(g)} \longrightarrow 2\ H_2O_{(l)}$$

What would happen if we have 2 mol of H_2 and 8 mol of O_2?

Would this change the amount of water that would be created?

NO! H_2 still runs out first.

A chemical reaction will stop once any one of the reactants runs out. This reactant is known as the limiting reagent. Any other reactant is an excess reagent.

Steps to stoichiometry with a limiting reagent:

1) Balance the chemical equation.
2) Assume one of the reagents as excess and one as limiting.
3) Calculate how much of the excess reagent is consumed if all of the limiting reagent is used. This will determine if the assumption was correct.
4) Complete stoichiometric calculation using the limiting reagent.

Ex) $N_{2(g)} + 3\, H_{2(g)} \longrightarrow 2\, NH_{3(g)}$

(16.4 g) (3.9 g) x grams

Assume H_2 is limiting

$$m_{N_2} = 3.9\,\cancel{g\,H_2} \times \frac{1\,\cancel{mol\,H_2}}{2.02\,\cancel{g}} \times \frac{1\,\cancel{mol\,N_2}}{3\,\cancel{mol\,H_2}} \times \frac{28.02\,g}{1\,\cancel{mol\,N_2}}$$

$$= 18\,g$$

$$\boxed{\therefore N_2 \text{ is limiting}}$$

Assume N_2 is limiting

$$m_{H_2} = 16.4\,g\,N_2 \times \frac{1\,mol\,N_2}{28.02\,g} \times \frac{3\,mol\,H_2}{1\,mol\,N_2} \times \frac{2.02\,g\,H_2}{1\,mol\,H_2}$$

$$= 3.55\,g$$

$\therefore N_2$ is limiting

$$m_{NH_3} = 16.4\,g\,N_2 \times \frac{1\,mol\,N_2}{28.02\,g} \times \frac{2\,mol\,NH_3}{1\,mol\,N_2} \times \frac{17.04\,g}{1\,mol\,NH_3}$$

$$= 19.9\,g$$

Ex) $Na_{(s)} + Cl_{2(g)} \longrightarrow NaCl_{(s)}$

(45.98 g) (142.0 g) x grams

$$m_{Cl_2} = \frac{45.98\,g}{1} \times \frac{1\,mol\,Na}{22.98\,g} \times \frac{1\,Cl}{2\,Na} \times \frac{70.86}{1\,mol\,Cl}$$

$$= 70.89\,g \quad \therefore Na \text{ is limiting}$$

$$m_{NaCl} = \frac{45.98\,\cancel{g\,Na}}{1} \times \frac{\cancel{1\,mol\,Na}}{22.98\,\cancel{g}} \times \frac{1\,mol\,NaCl}{1\,\cancel{mol\,Na}} \times \frac{58.44}{\cancel{1\,mol\,NaCl}}$$

$$= 116.9\,g \text{ of } NaCl$$

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Ex) 15.00 g of aluminum sulfide added to 10.00 g of water.

a) How much aluminum hydroxide is produced?

b) How much excess reagent is left?

$$Al_2S_{3(s)} + \quad H_2O_{(l)} \longrightarrow Al(OH)_{3(aq)} + \quad H_2S_{(aq)}$$